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SIM-69-Keypad

DESCRIPTION

SIM-69-Keypad is an add on board to attach to the SIM-69 Motherboard that provides for a 6 row by 4 column keypad and four controllable LEDs. The keypad board attaches to either of the VIA PortA and PortB connectors on the motherboard.

HARDWARE

The keypad uses many of the PortA and PortB lines from one of the VIAs on the motherboard. It has been designed to work around the existing port pins that are used from VIA 6522-2 PortB that can be used for bank switching the EEPROM on the motherboard. Thus the only logic on the keypad PCB is a single 74LS04, primarily used to drive the LEDs since the VIA output pins alone do not provide adequate current by themselves.

If the motherboard is using CMOS 6522 VIAs from Western Design Center, an optional resistor pack can be installed at RN1 to provide the keypad matrix columns pull up resistors. Older NMOS and CMOS 6522s contain an internal pullup that the WDC part lacks, so those parts will not require the resistor pack.

A spare 16 pin DIP package and a buzzer circuit are on the PCB but not used.

OPERATION

PortA pins PA0 through PA5 are configured as outputs, and all set hi. PortB pins PB4 through PB7 are configured as inputs. This forms the 6x4 matrix of keys. PortA pins can be individually set low and the PortB pins read. If a low is seen on a PortB pin, a key in the matrix has been pressed. 1N4148 signal diodes are used to eliminate ghosting.

The LEDs can be controlled individually by the remaining unused port pins on the VIA; PA6, PA7, CA2 and CB2. These would be configured as outputs and set hi to light an LED..

PARTS

The keypad PCB was designed with the Keebio Nyquist key plate in mind, to keep the cherry MX switches aligned properly. This plate is optional but if not using a plate, you may notice your keycaps are not uniformly spaced between rows and columns, as each switch may be rotated slightly when soldered to the PCB. The lower right key is a 2U keycap for an enter key, and it required a 2U stabilizer for smooth operation. A PCB or “screw in” stabilizer is used.

Since the Keebio Nyquist plate is really designed for a landscape layout, and we are using it in a portrait layout, the cherry MX switches are laid out 90 degrees rotated from what would be considered “normal” on the PCB. This allows the switches to snap into the plate snugly. There is no operational or physical difference due to the rotation as the keys switches are square, and the stems are centered in either orientation.

Signature Plastics sells several keycap sets, including a set that contains the full Hexadecimal keys, 0-9 and A-F. Check the Bill Of Materials list for suggestions on where to get the “specialty” parts.